WW-S-2739
October 24, 1989
SUPERSEDING
MIL-S-16293G
18 January 1980

FEDERAL SPECIFICATION

STRAINERS, SEDIMENT: PIPELINE, WATER, AIR, GAS, OIL, OR STEAM

This Specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers removable basket-and-screen-strainers of various sizes for use in water, air, gas, oil, or steam pipe lines, as specified (see 6.2 and 6.4), to remove dirt, scale particles, and other foreign matter.

1.2 Classification. Strainers will be of the following types, classes, styles, and sizes as specified (see 6.2):

Type I - Screwed connections.
Type II - Flanged connections.

Class 125 - 125 pounds per square inch gage (psig) pressure rating.
Class 250 - 250 psig pressure rating.

Style I - I pattern.
Style L - L pattern.
Style S - S pattern.
Style T - T pattern.
Style Y - Y pattern.

1.2.1 Sizes. Types I and II strainers will be available in the following sizes (iron pipe size in inches):

Type I - 3/8, 1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, and 3.
Type II - 2, 2-1/2, 3, 4, 5, 6, and 8.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4730

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1.2.2 Part number. The military part numbers for items in this specification will be identified as shown in 6.5.

2. APPLICABLE DOCUMENTS

2.1 Government document.

2.1.1 Specification. The following specification forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Military Specification

MIL-V-3 - Valves, Fittings, and Flanges (Except for Systems Indicated Herein); Packaging of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publication. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are Department of Defense (DoD) adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which is current on the date of the solicitation (see 6.2).

American National Standards Institute, Inc. (ANSI)

ANSI B16.1 - Cast-Iron Pipe Flanges and Flanged Fittings, Class 25, 125, 150, and 800 Pounds.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Non-Government standards and other publications are normally available from the organizations which prepare or that distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description.

3.1.1 Type I. Type I strainers shall have threaded connections, or be provided with adaptors for threaded connections, and have removable screens, and be of the class and style required. When specified (see 6.2), the strainers shall have rotating screens or scraper blades, manual or motor operated, for the purpose of cleaning screens without shutting down operation of the strainer. Unless specified as a twin or duplex type (see 6.2), strainers for placement in the inlets of pressure-reducing valves, traps, or heating system pumps shall be of the single type screen.

3.1.2 Type II. Type II strainers shall have flanged connections, be of the center or peripheral influent type of the class and style specified, and have removable straining elements. When specified (see 6.2), they shall have rotating screens or scraper blades, manual or motor operated, for purpose of cleaning screens without shutting down operation of the strainer. The number of straining elements shall be determined by the application in accordance with the best industrial standards.

3.2 Standard commercial product. The strainers shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the strainers being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.3).

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

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3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Pressure-temperature ratings. Pressure-temperature ratings shall be not less than the following:

a. Class 125 - 125 psig saturated steam pressure; and 175 psig at 150° Fahrenheit (°F).
b. Class 250 - 250 psig saturated steam pressure; and 400 psig at 150°F.

3.7 Hydrostatic test pressure. All strainers shall be capable of withstanding a hydrostatic shell test pressure equal to twice the primary steam service pressure (see 3.6).

3.8 Styles.

3.8.1 Styles I and L. Styles I and L strainers shall have the access opening in line with the filter element.

3.8.2 Styles S and T. Nonscraper type, styles S and T strainers shall be designed for positioning the element vertically within the body. Scraper type strainers may have the element positioned at an angle within the body. Unless otherwise specified (see 6.2), the access opening shall be located at the top of the body. A clean-out connection shall be tapped in a boss on the bottom of the body.

3.8.3 Style Y. The branch leg of the Y pattern shall be inclined 30 to 65° from the straight run. The strainer element shall be located on the branch leg of the Y pattern and shall be removable through the end of the branch leg. The clean-out connection shall be tapped in the access cap or cover.

3.9 Construction.

3.9.1 Strainer bodies. Unless otherwise specified (see 6.2), the bodies shall be malleable iron, or cast iron. The bodies shall be so designed that the flow is into the inlet and either to the inside or outside of the strainer element as specified (see 6.2), then through the strainer element to the outlet of the body and into and through the outlet connection.

3.9.2 Connections. All type I strainer connections shall be tapped with American National Taper Pipe threads. All type II strainer connections shall be flanged and drilled for bolts in accordance with ANSI B16.1.

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3.9.3 Strainer element detail. The element shall be formed into a cylinder or frustrum of a cone with one or both ends open. The element shall be properly reinforced to prevent deformation resulting in by-passing sediment particles. All joints in the element shall be joined by silver brazing, soldering, riveting, welding, or crimping. Elements shall be constructed of perforated sheet metal or of woven wire mesh as specified (see 6.2).

3.9.4 Strainer element material. Strainer elements shall be of the following materials for the service specified (see 6.2):

a. Steam, water, gas or air - Brass, bronze, nickel-copper-alloy, or stainless steel sheet metal; and phosphor-bronze, nickel, copper alloy or stainless steel wire mesh.
b. Oil service - Iron or steel and the materials included in subparagraph a. of 3.9.4.

3.9.5 Size of openings. Unless otherwise specified (see 6.2), elements shall have wire mesh openings, or perforations of 0.014 inch minimum for steam, 0.031 inch minimum for steam mixed with condensate, and 0.042 inch minimum for water or low-viscosity oil.

3.9.6 Ratio of net strainer area to pipe area. Unless otherwise specified (see 6.2), the ratio of the net effective strainer area to the cross-sectional flow area of standard weight pipe of the same nominal size as the strainer shall be not less than 2.50 to 1. When mesh and perforated metal are used together, the open area should be the result of multiplying the open areas of each together.

3.9.7 Access cover. Unless otherwise specified (see 6.2), the access cover shall be cast iron or malleable iron and shall be secured to the strainer body by means of through bolts, hinged bolts, yokes, cap screws, or studs of appropriate size and number. Whether screwed or bolted, the access cover may have a female pipe thread for a blow-down connection, in which shall be installed a brass pipe plug. The plug shall be tight under test and service conditions. Strainers of 3-inch pipe size and smaller may have a threaded access cover with either pipe threads and no gasket, or straight threads with a gasket.

3.9.8 Cover gasket. Whether the access cover is screwed or bolted, a gasket shall be installed between cover and body to insure cover tightness under test and service conditions. The gasket shall be composed of a material suitable for the intended application and use, as specified (see 6.2). For type 1, style Y, screwed connection strainers in size 2 inch and under, a gasket need not be furnished when the access cover is secured with tapered pipe threads.

3.9.9 Bottom drain plug. If supplied, the plug shall be brass.

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3.10 Marking.

3.10.1 Service marking. Each strainer shall bear the following service markings, cast integral with, stamped or otherwise permanently marked upon the body:

a. "125" as pressure identification for class 125 strainers.
b. "250" as pressure identification for class 250 strainers.
c. An arrow or other appropriate mark to indicate flow direction.

3.10.2 Identification marking. Equipment, assemblies, and parts shall be marked for identification. The nomenclature of the item shall be "Strainer, Sediment". When specified (see 6.2), the strainer shall include the Federal Stock Number.

3.11 Spare parts and maintenance tools. When specified (see 6.2), spare parts and maintenance tools shall be furnished.

3.12 Workmanship. The strainers shall be clean, free from scale, smooth, round, straight, and of proper dimensions. Also, the strainers shall be free from injurious grooving, indentations, racks, flaws, slivers, or other harmful imperfections which would interfere with use of the strainer in the application for which it is intended.

3.12.1 Metal fabrication. Metal used in fabrication of equipment shall be free from kinks. Material shall be straightened by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Flame cutting, with use of a tip suitable for the thickness of metal, may be employed instead of shearing or sawing, provided that exposed edges are smoothly made. All bends shall be made by controlled means to ensure uniformity of size and shape. Precautions shall be taken to avoid overheating and heated metal shall be allowed to cool slowly.

3.12.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. All bolts, nuts, and screws shall be tight.

3.12.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.12.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

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3.12.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

3.12.6 Machine work. Tolerances for contact and bearing surfaces shall conform to standards prevailing among manufacturers normally producing strainers.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).
c. Packaging inspection (see 4.7).

4.2.1 First article inspection. First article inspection shall be performed on one complete strainer when a first article sample is required (see 3.3 and 6.2). This inspection shall include the examination of 4.5 and the tests of 4.6. The first article may be a standard production item from the contractor's current inventory, provided the strainers meet the requirements of this specification and is representative of the design, construction, and manufacturing technique applicable to the remaining strainers to be furnished under the contract.

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4.2.2 Quality conformance inspection. Quality conformance inspection shall be performed on the sample strainer selected in accordance with 4.2.1. This inspection shall include the examination of 4.5 and the tests of 4.6.

4.3 Inspection lot. All units of the same types, classes, styles, and size offered to the Government at one time, shall be considered a lot for purpose of inspection. The sample unit shall be one complete strainer.

4.4 Sampling.

4.4.1 Sampling for tests. (See 6.4.1.)

4.4.2 Sampling for examination. (See 6.4.2.)

4.5 Examination. Each strainer selected in accordance with 4.4.1 shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Non-compliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.6. Tests. Each production unit shall be tested as specified in 4.6.1 and 4.6.2. Failure to pass any test shall constitute cause for rejection.

4.6.1 Pressure-temperature. Subject strainer to the applicable pressure-temperature ratings as specified in 3.6. Nonconformance to the requirements specified in 3.6 shall constitute failure of this test.

4.6.2 Hydrostatic pressure. Subject strainer to a hydrostatic pressure as specified in 3.7. Any seepage or leakage through the casting or at any connection or joint shall constitute failure of this test.

4.7 Packaging inspection. The preservation, packaging, packing, and marking of the strainers shall be inspected to determine conformance with the applicable requirements of section 5 of this specification.

5. PACKAGING

5.1. Preservation, packaging, packing, and marking. The strainer shall be preserved, packaged, packed, and marked in accordance with MIL-V-3. The level of preservation and packaging and level of packing shall be as specified (see 6.2).

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The strainers covered by this specification are intended for general service to protect regulating valves, orifices, pumps, and other piping equipment from damage or faulty operation which would result from entrained sediment particles.

6.1.1 Perforations and mesh. The following ranges of perforation sizes and wire mesh numbers are acceptable.

a. Perforation diameters, inches (approximate):

0.020 to 0.023
0.027 to 0.033
0.040 to 0.050
0.057 to 0.063
0.090 to 0.105
0.120 to 0.130

b. Mesh numbers - 20, 40, 60, 80, and 100.

6.1.2 Ratio of net strainer area to pipe area. The ratio of the net effective strainer area to the area of the pipe in an indirect index of the resistance to flow through the strainer. The ratio cited in 3.9.6 is a minimum value for the range of sizes covered by this specification. Many designs of strainers provide for higher area-ratios. The area-ratio is dependent upon the size of opening and the total surface area of the element, and varies with the size of the strainer. If a higher ratio is required, manufacturer's literature should be consulted to ascertain availability of the required area-ratio,.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).
c. Type, style, class, and size required (see 1.1, 1.2, 3.1.1, and 3.1.2).
d. Whether a twin or duplex arrangement is required (see 3.1.1).
e. When a rotating screen or scraper blade is required (see 3.1.1.and 3.1.2).
f. When first article is required for inspection and approval (see 3.3, 4.2.1, and 6.3).

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g. Location of access opening if different (see 3.8.2).
h. Material for body assembly required, and whether flow will be to the inside or the outside of the strainer element (see 3.9.1).
i. Whether strainer element is to be wire mesh or perforated metal (see 3.9.3).
j. Strainer element material required (see 3.9.4).
k. Size of strainer element openings or net effective area ratio required, if other than specified (see 3.9.5 and 3.9.6).
l. Material for access cover, if other than specified (see 3.9.7).
m. Material for access cover gasket (see 3.9.8).
n. When the Federal Stock Number is to be included (see 3.10.2).
o. When spare parts and maintenance tools are required (see 3.11).
p. When first article inspection does not include all examination and tests applicable requirements of Section 3 (see 4.2.1).
q. Level of preservation and packaging and level of packing required (see 5.1).

6.3 First article. When a first article inspection is required, the item will be tested and should be a first production item, or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one. The contracting officer should include specific instructions in acquisition documents, regarding arrangements for examination, test, and approval of the first article.

6.4 Sampling procedures.

6.4.1 Sampling for tests. Recommended inspection level is II and Acceptable Quality Level (AQL) is 2.5 percent defective (see 4.4.1).

6.4.2 Sampling for examination. Recommended inspection level is II and the AQL of 2.5 percent defective (see 4.4.2).

6.5 Definitive federal specification part number. The Federal specification part number is a definitive part number which corresponds to the type, class, size, and style of strainers covered by this specification. The Federal specification number, the type and class code letter, and the size and style code numbers are combined to form a definitive Federal specification part number.

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6.5.1 Cataloging data. For cataloging purposes, part numbers for the strainers are assigned as follows:

	WW-S-2739	X	XX
Federal specification number			
Type and class code letter			
Size and style code number			

6.5.2 Type and class. The type and class of strainer (see 1.2) are identified by a single letter (see table I).

TABLE I. Code letter to type and class.

	Type I	Type II
Class 125	A	B
Class 250	C	D

6.5.3 Size and style. The size and style of strainer (see 1.2) are identified by a double digit number (see table II).

TABLE II. Code number to size and style.

	STYLE				
Size	I	L	S	T	Y
3/8	01	02	03	04	05
1/2	06	07	08	09	10
3/4	11	12	13	14	15
1	16	17	18	19	20
1-1/4	21	22	23	24	25
1-1/2	26	27	28	29	30
2	31	32	33	34	35
2-1/2	36	37	38	39	40
3	41	42	43	44	45
4	46	47	48	49	50
5	51	52	53	54	55
6	56	57	58	59	60
8	61	62	63	64	65

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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MILITARY INTERESTS:

Custodians:

Army - ME
Navy - YD
Air Force - 99

Review Activities:

Air Force - 82
DLA - CS

User Activities:

Army - CE
Navy - CG, MC

COORDINATING ACTIVITIES:
CIVIL AGENCY

GSA - FSS

PREPARING ACTIVITY:

Navy - YD

Project 4730-1158

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
WW-S-2739 -	STRAINERS, SEDIMENT: PIPELINE, WATER, AIR, GAS, OIL, OR STEAM

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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